

Various Sensor Types Table 2

Sensor Type	Company Name
RGB	Canon Powershot SX 540
	Olympus PEN E-PM1
	Sony Nex-7 / ILCE
	Ricoh GR3
	Sony Alpha 7r
Multi-spectral	MicaSense RedEdge
	MCA camera
	Mini MCA
	Mirco MCA
	Parraot Sequoia
	InGaAs
	Tetracam ADC Lite
	Tetracam ADC Micro
Laser	Veldyne VLP-16
	Hokuyo UTM-30LX
	Nikon Forestry Pro II
	Rangefinder/ Hysometerm
Thermal	DJI Zenmuse XT
	Xenics Bobcat 640 GigE SWIR
	Thermoteknix MicroCAM Integrator
	FLIR Vue Pro R
Multi-scanner	RapidSCAN CS-45

Vegetation Index (NDVI) use detailed colour information to determine plant health. Drones monitor crop health regularly as compared to satellite imagery which is costly and not as effective in all weather conditions. As drones fly close to fields, cloud cover and poor light conditions matter less than when using satellite imaging. Satellite imaging may be accurate to a metre, but drone imaging is accurate to a millimetre. This means that after planting, areas with gaps can be spotted and replanted, and disease or pest problems can be detected and treated for right away.

Field condition monitoring. Drones can help monitor the health of soil and field conditions and detect all sorts of field irregularities.

Field spraying. Drone sprayers can navigate hard-to-reach areas such as steep tea fields at high elevations. Drone sprayers save workers from having to navigate fields with backpack sprayers, which can be hazardous to their health.

Security. Drone cameras can pro-

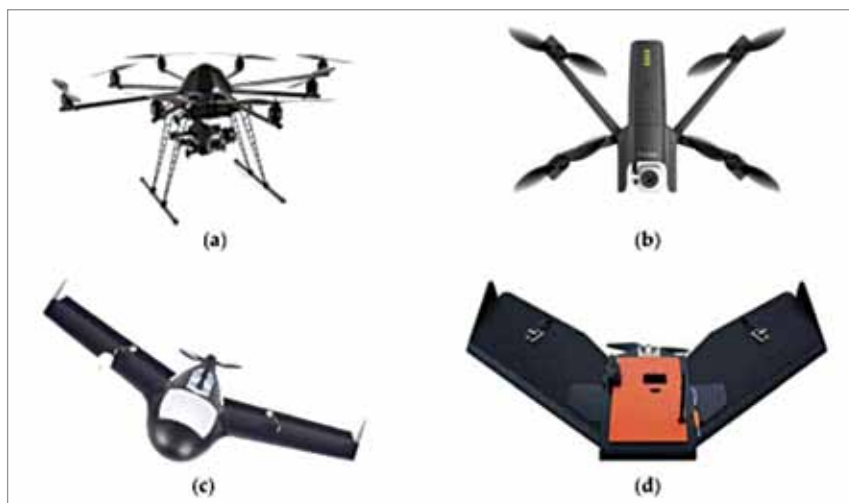


Fig. 3: Commercial drones used for precision agriculture purposes: (a) 8-rotor MK Okto XL 2 of HiSystems GmbH/Mikrokooper.de, (b) quad-rotor Parrot Anafi, (c) Gatewing X100, and (d) Tuffwing Mapper. P



Fig. 4: DJI Agras MG-1

Technical Specifications of DJI Agras MG-1

Table 3

Specification Type	Values
Frame Arm Length	625mm
Dimensions	1471mm x 1471mm x 482mm (arm unfolded, without propellers) 780mm x 780mm x 482mm (arm folded)
Max Battery Size	151mm x 195mm x 70mm
Total Weight (without batteries)	8.8kg
Max Thrust-Weight Ratio	1.81 (with 22.5kg takeoff weight)
Hovering Power Consumption	3250W (with 22.5kg takeoff weight)
Recommended Operating Temperature	0 to 40°C
Remote Controller - Model	GL690B (Japan only, GL658C)
Built-in Battery	6000mAh, 2S LiPo

vide an overview of farm operations and equipment in the field, throughout the day, to ensure operations are running smoothly. Security drones can

be deployed to monitor fencing and perimeters of more valuable crops like cannabis instead of employing more security personnel.